Abstention and Routing: Experiments and Findings

Two experiments on the substitutability committee, aimed at the deployment target (**agree with the LLM judge 95-99% of the time while sending as few rows as possible to the LLM**). All results are on the **721-row held-out test split**; the committee is fit on the 222-row validation split. Target labels are the LLM judge's `substitutable` decisions (`gpt-oss-120b`).

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1. Methodology

Experiment 1 - Abstention (weak-signal dead-zone)

Each program's normalized score is turned into a vote with a **dead-zone of half-width `band`** around its tuned decision threshold theta: if the score lies in `[theta - band, theta + band]` the program **abstains** (votes "no opinion") rather than forcing a 0/1. We sweep `band in {0, 0.05, 0.10, 0.15, 0.20}`, refit the full pipeline (threshold tuning -> >=0.50 filter -> top-k selection -> Snorkel aggregation) at each band, and report **accuracy on committed rows** and **coverage** (fraction of rows the committee still decides). This is the paper's per-program abstention margin (§3.3).

Experiment 2 - Routing (paper-exact)

On the base committee (`band = 0`) we compute two **deterministic** confidence signals from the committee output - **no LLM call is used to decide routing**: - **Aggregator-posterior margin** `|P(substitutable) - 0.5|` (close to 0 = unsure). - **Vote variance** - disagreement among the selected programs' votes (high = unsure).

We sweep an escalation threshold from 0->100%, routing the least-confident rows first. We compare both signals against an **oracle** (routes the committee's wrong rows first - the best-possible policy) and a **random** baseline, and report the escalation rate needed to reach 0.95 and 0.99 agreement. (The paper's length-based baselines don't apply to short food strings and are omitted.)

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2. Assumptions (important)

- **Routed rows are counted correct by identity, and no LLM was actually called.** Our eval target *is* the LLM judge's label; "escalate" means "ask that same judge," which returns the exact stored label. Since the judge runs at temperature 0, re-asking is deterministic, so escalated rows agree with the target **by definition** (not by approximation). This is why the 100%-escalation endpoint is exactly 1.0.
- **This differs from the PAJAMA paper.** There, the escalation LLM and the ground truth are *different*, so their pure-LLM endpoint is the judge's own accuracy (< 1.0) and the hybrid can *exceed* it. In our "reproduce-the-LLM-judge" framing the committee can only *approach* the LLM, never beat it - which matches the Direct Supply metric ("agree with the LLM").
- **Against human truth the assumption breaks:** an escalated row gets the LLM's answer, which is only ~0.91 vs human, so a human-targeted curve would cap at ~0.91.
- Committee base accuracy on test = **0.749** (wrong fraction 0.251); labels precomputed.

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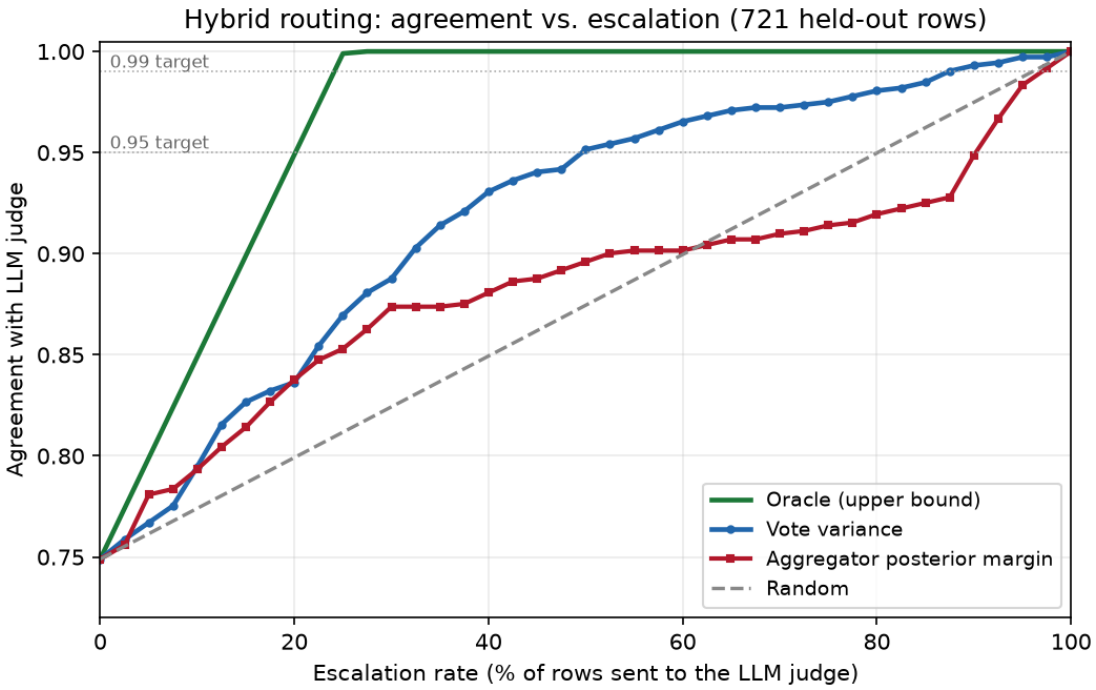
3. Results

Experiment 1 - Abstention

band	best_k	coverage	accuracy (committed)
0.00	12	1.000	0.749
0.05	2	0.863	**0.794**
0.10	20	1.000	0.723
0.15	9	0.943	0.740
0.20	18	0.964	0.742

Not a clean win. The only gain is at `band = 0.05` (0.749 -> 0.794 at 86% coverage), and `best_k` swings erratically (12 -> 2 -> 20 -> 9 -> 18). The tell: the dead-zone is applied *before* threshold tuning, filtering, selection, and Snorkel, so it **destabilizes committee selection** instead of producing a smooth accuracy-vs-coverage trade. The global dead-zone is therefore **not a reliable de-noising lever** here.

Experiment 2 - Routing



escalation	oracle	vote variance	posterior margin	random
0%	0.749	0.749	0.749	0.749
20%	0.949	0.836	0.838	0.799
30%	1.000	0.888	0.874	0.824
50%	1.000	**0.951**	0.896	0.874
85%	1.000	0.985	0.925	0.962
100%	1.000	1.000	1.000	1.000

Escalation rate needed to hit the target:

target	oracle	vote variance	posterior margin	random
0.95	**22.5%**	**50.0%**	92.5%	82.5%
0.99	25.0%	87.5%	97.5%	97.5%

Findings. 1. **Routing works; vote variance is the usable signal.** It reaches 0.95 at **~50%** escalation vs random's 82.5% - roughly a **2x LLM-cost saving** at the 95% target, with no LLM calls to decide routing. 2. **Posterior margin is a poor signal (~ random).** The Snorkel posterior is **miscalibrated** on this noisy committee, so `|P - 0.5|` barely ranks errors - a direct symptom of the weak programs identified earlier. 3. **Large oracle gap.** Oracle hits 0.95 at **~22.5%**; the best real signal needs ~50% - i.e. the deployable router spends roughly **2x more** LLM calls than theoretically necessary. That gap is the core routing research opportunity (RQ2).

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4. Suggestions to improve / reach the target

Ordered by expected payoff toward "95-99% agreement at low escalation":

1. **Learn the router (close the oracle gap - RQ2).** Replace the single-signal threshold with a small model (logistic regression / gradient-boosted trees) trained on the validation split over the **full committee state** - every program's vote, score magnitude, abstain flag, the posterior, and vote variance - to predict "will the committee be wrong here." This should move the curve from vote-variance (~50% @ 0.95) toward the oracle (~22.5%), still with **zero per-row LLM cost**. Highest-leverage single change.
2. **De-noise the committee first.** Prune the ~3 coin-flip programs (test AUC ~ 0.5) and feed the aggregator **continuous calibrated scores** instead of a single-theta vote. This both raises base accuracy (fewer errors to escalate -> lower escalation at any target) and should **rescue the posterior signal**, which is currently wasted.
3. **Calibrate the aggregator posterior (RQ4).** Fit Platt/isotonic calibration on val so `P(substitutable)` is trustworthy; add input-dependent program weighting and correlated-error handling (CARE). A calibrated posterior makes margin routing competitive with vote variance, and the two can be combined.
4. **Raise base committee accuracy (RQ3 / rubric redesign).** Repair the over-strict prep/preservation programs on their failure rows, and give the weak rubrics (specificity/modifier/processing) a **core-food-identity backbone**. Every point of base accuracy shifts the whole routing curve up, lowering the escalation needed for 95-99%.
5. **Abstention done right.** Drop the global dead-zone (it destabilizes selection). Instead abstain **program-internally** ("no opinion when nothing relevant is detected") or apply abstention **only at inference** (post-fit), and route the abstained rows - so abstention feeds the router rather than fighting the fit.
6. **State the deployable contract now.** Even today the system supports *"escalate 50% -> 0.95 agreement with the LLM"* (~2x cost saving); closing the oracle gap targets *"escalate ~22% -> 0.95"* (~4.4x saving). Report the contract as the headline deliverable.
7. **Grow the human-labeled set.** Needed to confirm that 95-99% agreement-with-the-LLM corresponds to acceptable *human* quality, and to unlock the regime where the committee can actually **beat** the LLM (only visible against human ground truth).